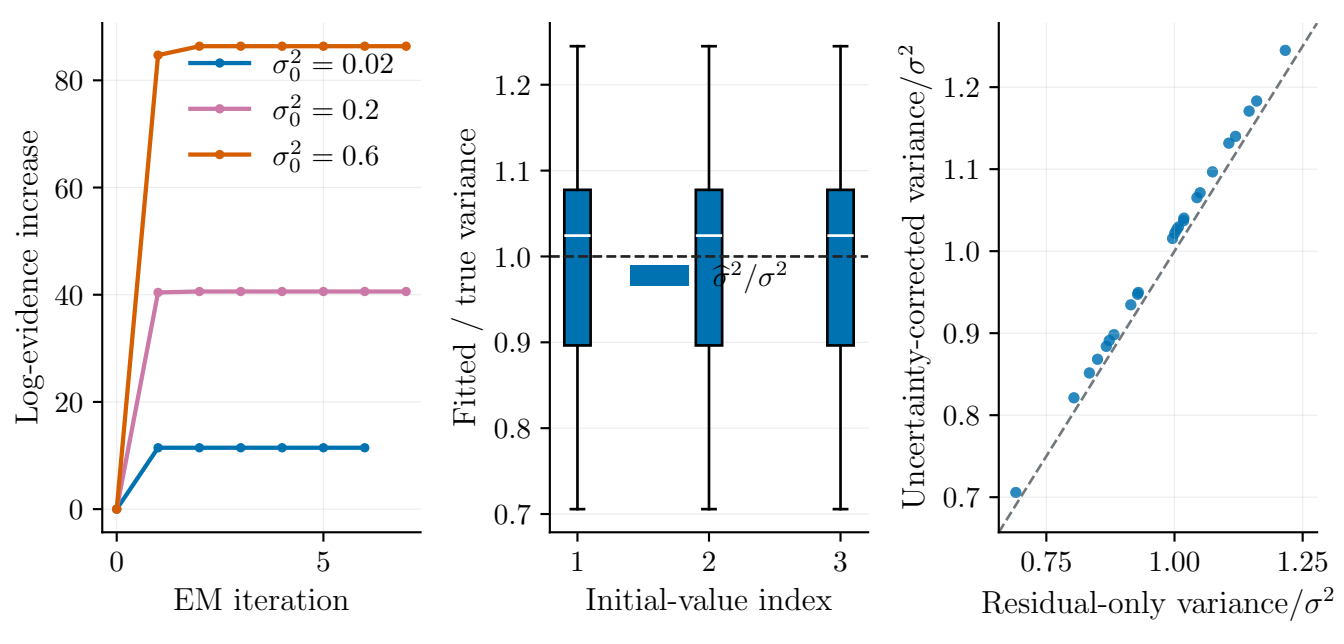
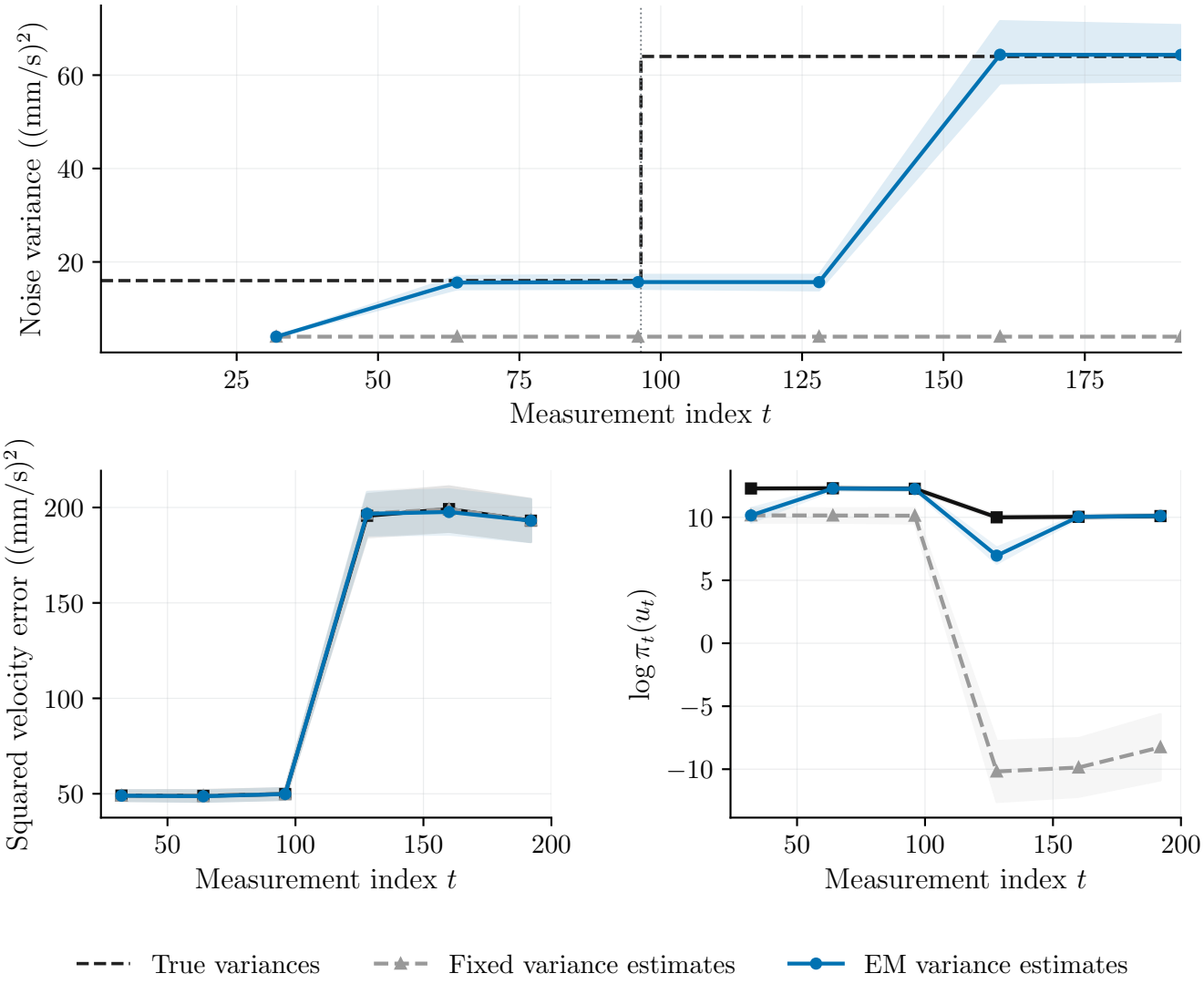
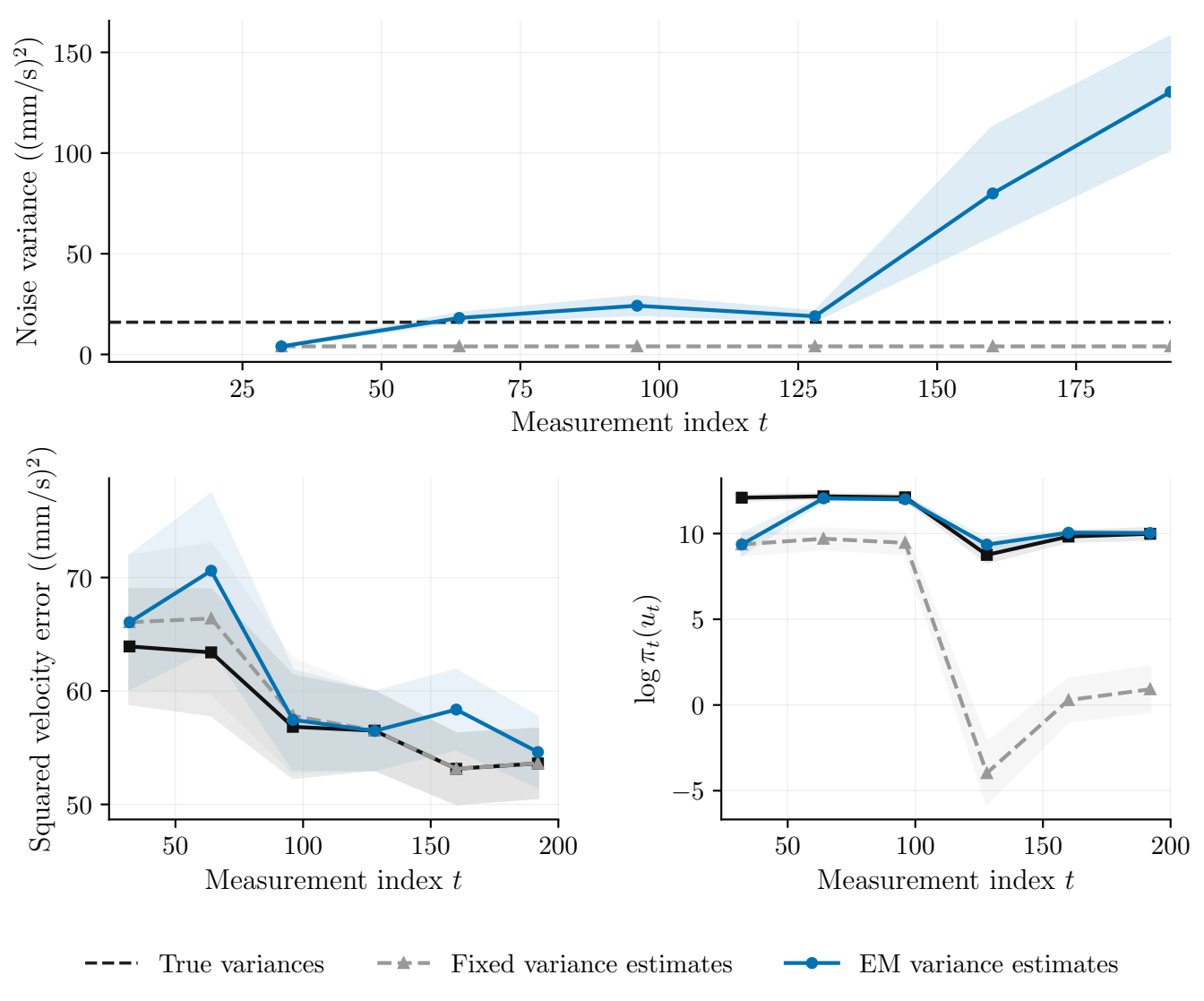
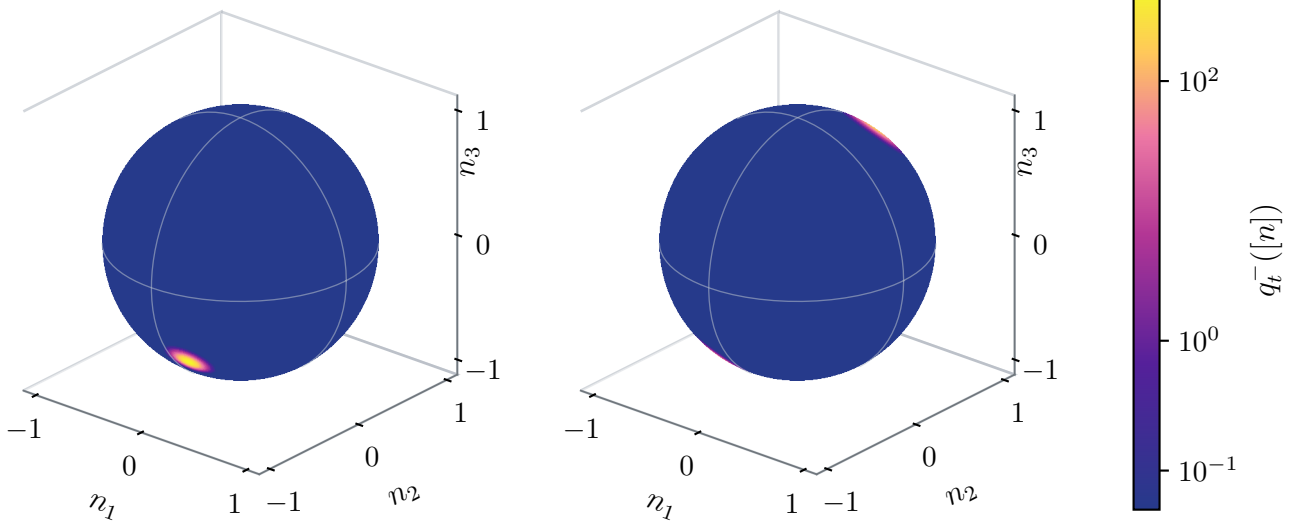
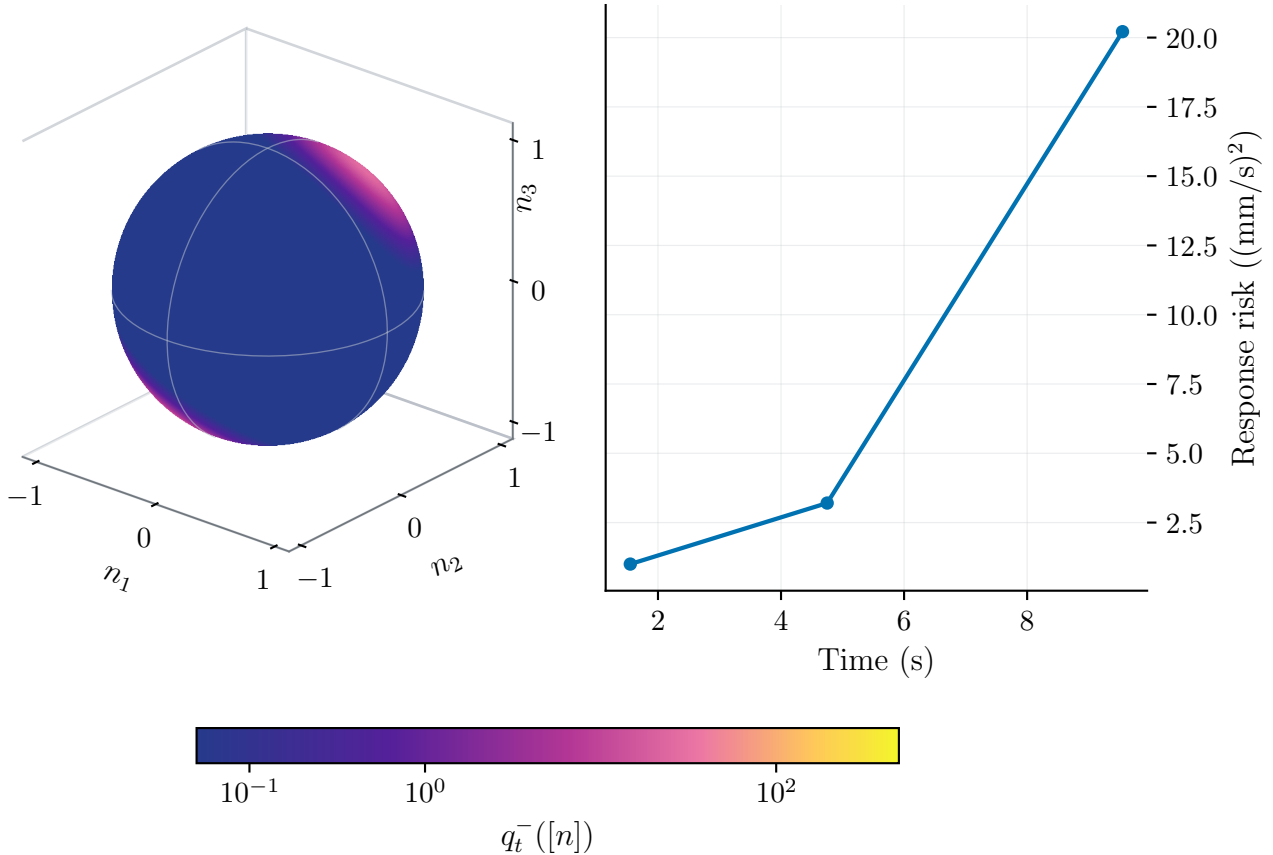
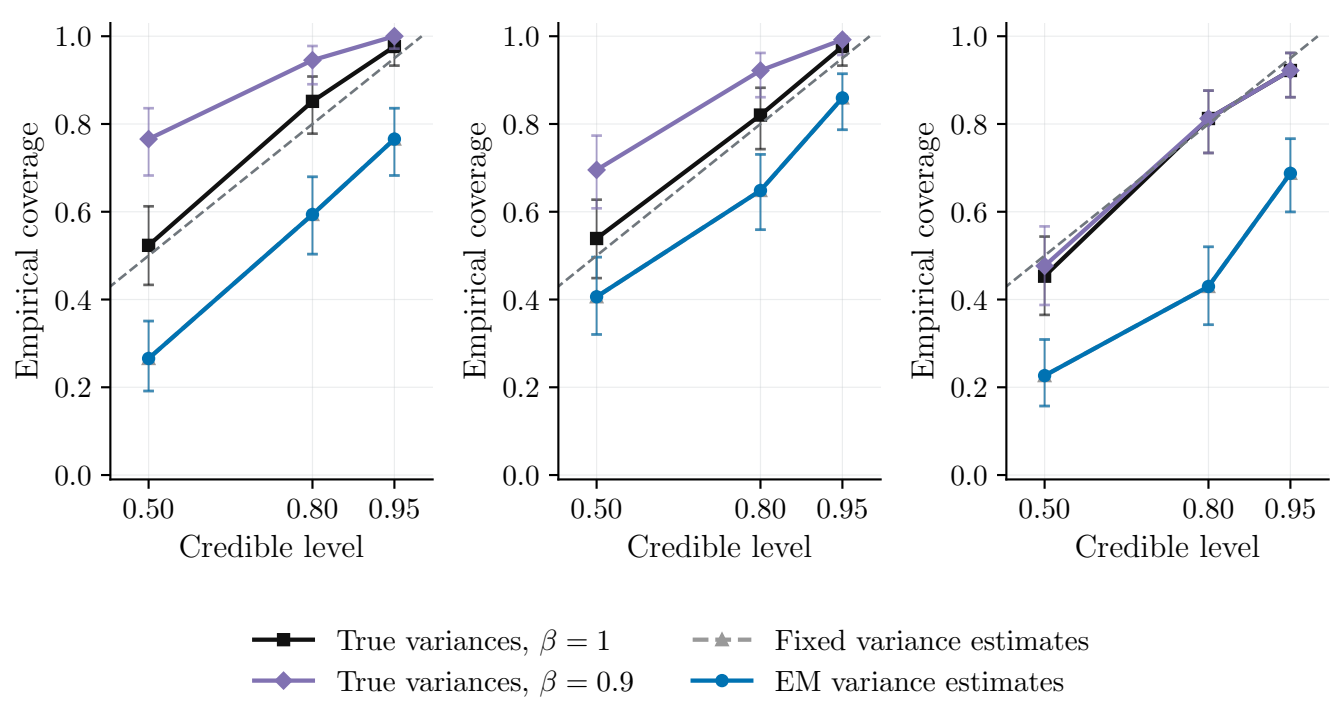

Log-evidence increase
EM iteration
$\sigma_0^2 = 0.02$
$\sigma_0^2 = 0.2$
$\sigma_0^2 = 0.6$
Fitted / true variance
Initial-value index
$\hat{\sigma}^2/\sigma^2$
Uncertainty-corrected variance/σ^2
Residual-only variance/σ^2

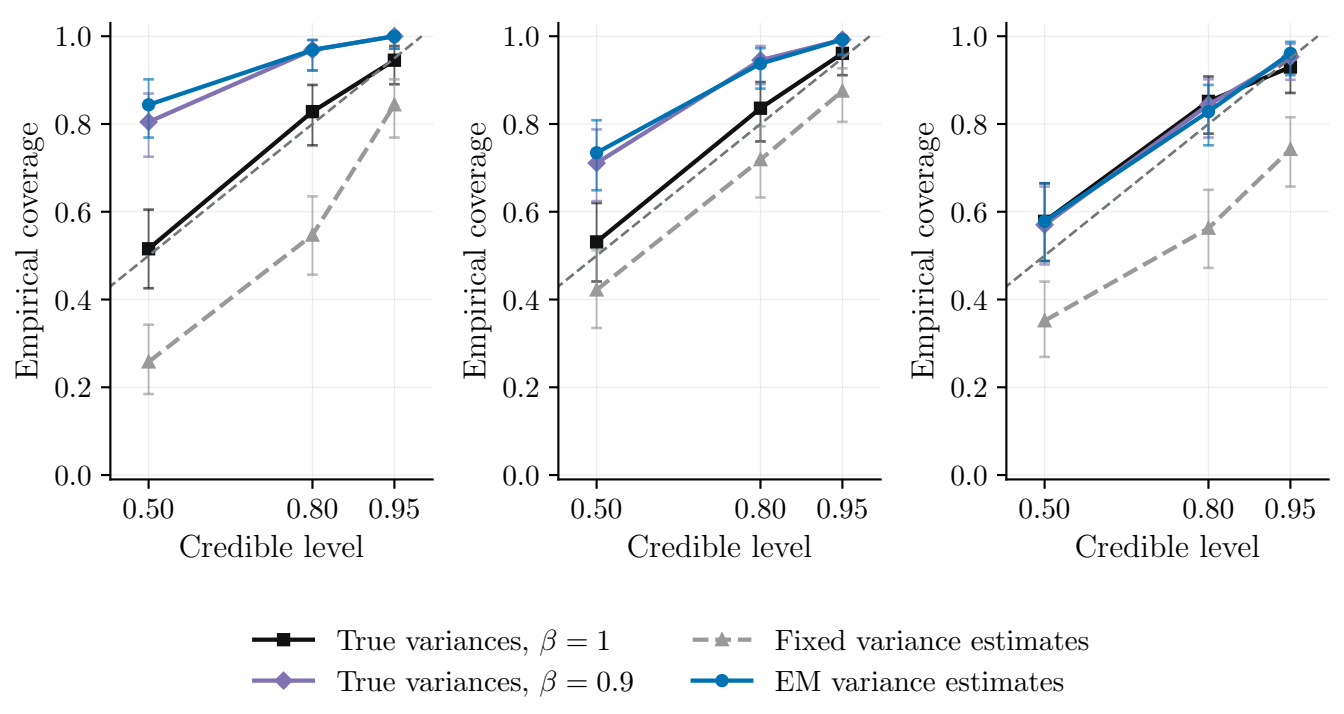

Noise variance ((mm/s)^2)
20
40
60
25
50
75
100
125
150
175
Measurement index t
Squared velocity error ((mm/s)^2)
50
100
150
200
50
100
150
200
Measurement index t
$\log \pi_t(u_t)$
10
5
0
−5
−10
50
100
150
200
Measurement index t
True variances
Fixed variance estimates
EM variance estimates

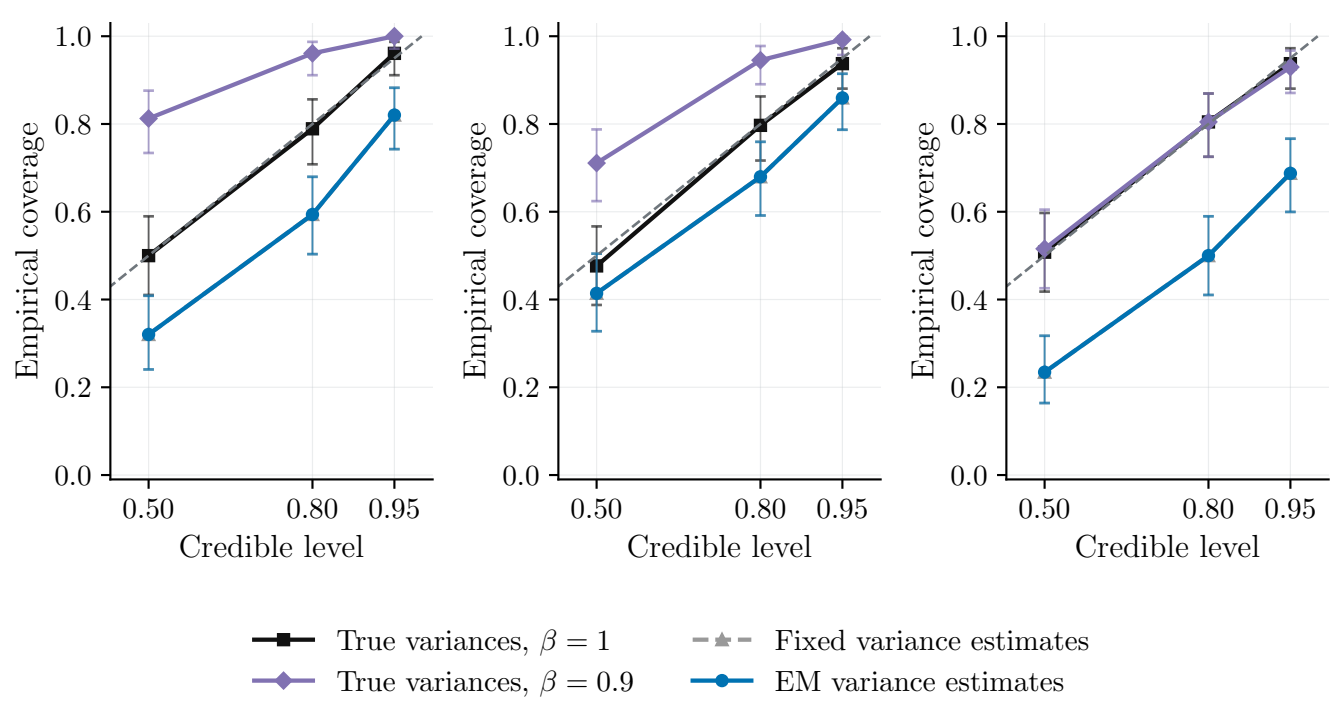

Noise variance ((mm/s)^2)
150
100
50
0
25
50
75
100
125
150
175
Measurement index t
Squared velocity error ((mm/s)^2)
70
60
50
50
100
150
200
Measurement index t
$\log \pi_t(u_t)$
10
5
0
−5
50
100
150
200
Measurement index t
True variances
Fixed variance estimates
EM variance estimates

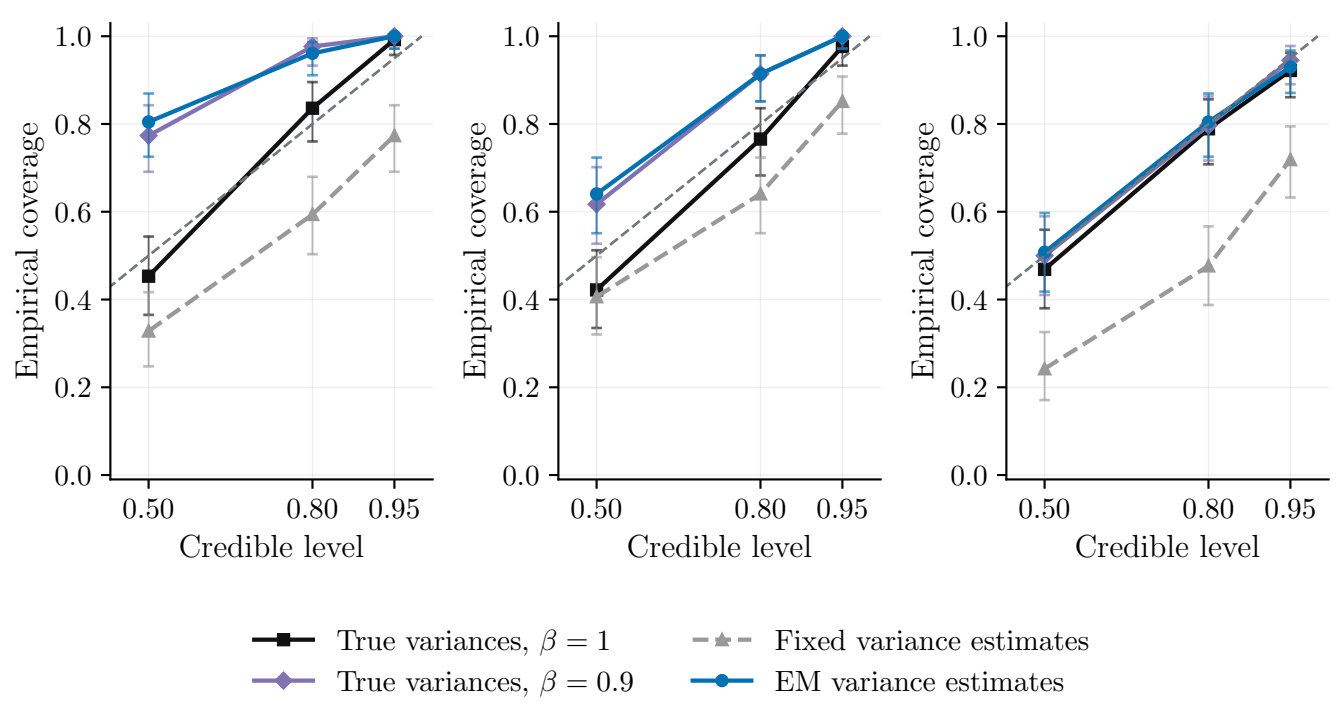

1
0
−1
n_3
−1
0
1
n_1
1
0
−1
n_2
10^2
10^0
10^{-1}
$q_t^-([n])$

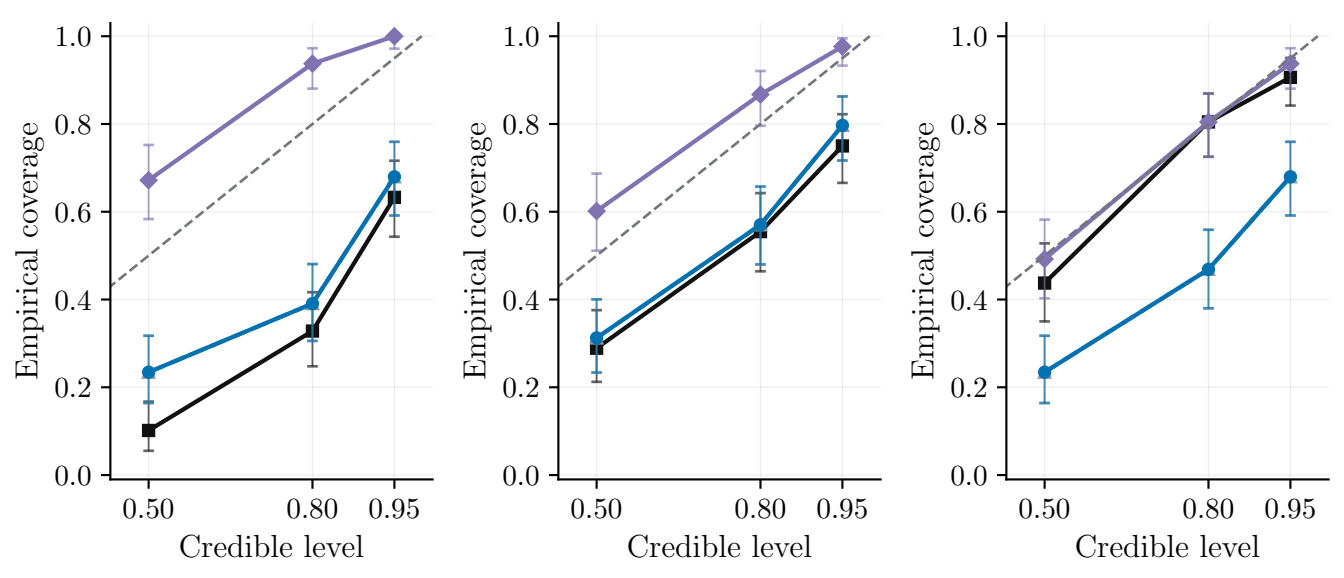

n_3
1
0
−1
−1
0
1
n_1
−1
0
1
n_2
Response risk ((mm/s)²)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
2
4
6
8
Time (s)
10^{-1}
10^{0}
10^{2}
$q_t^-([n])$

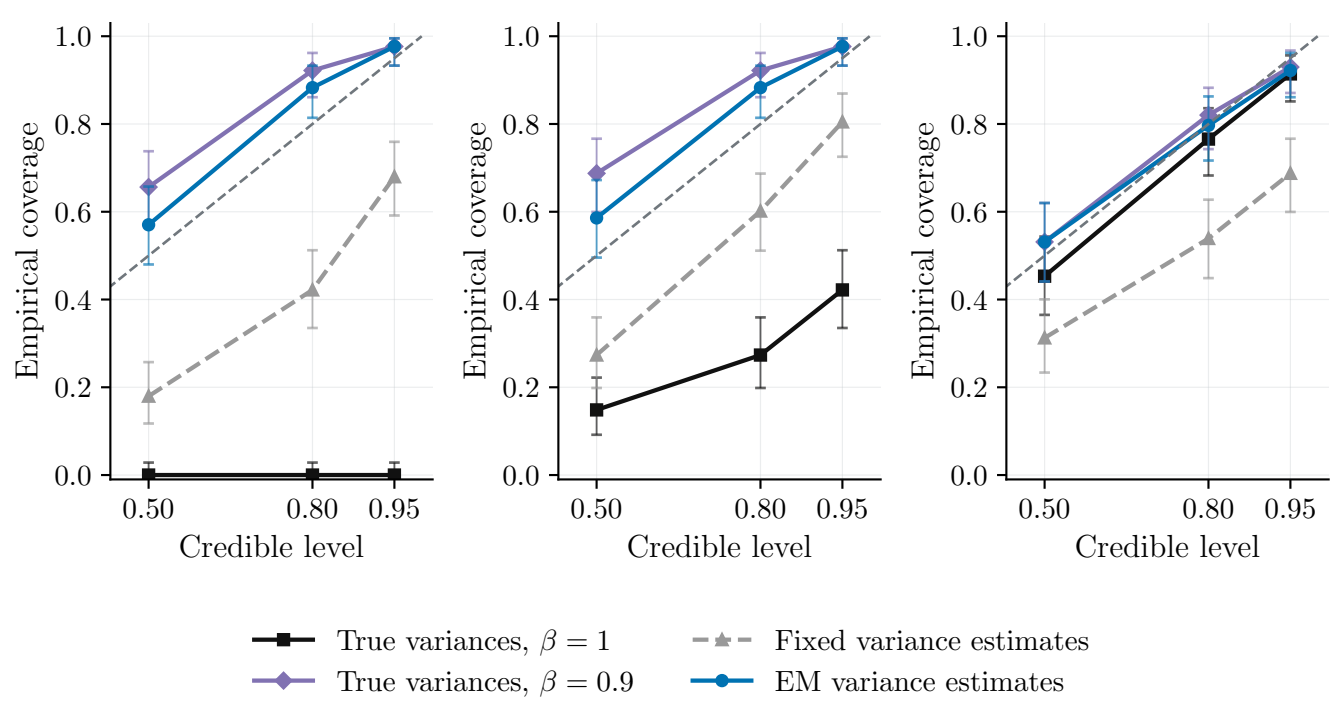

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

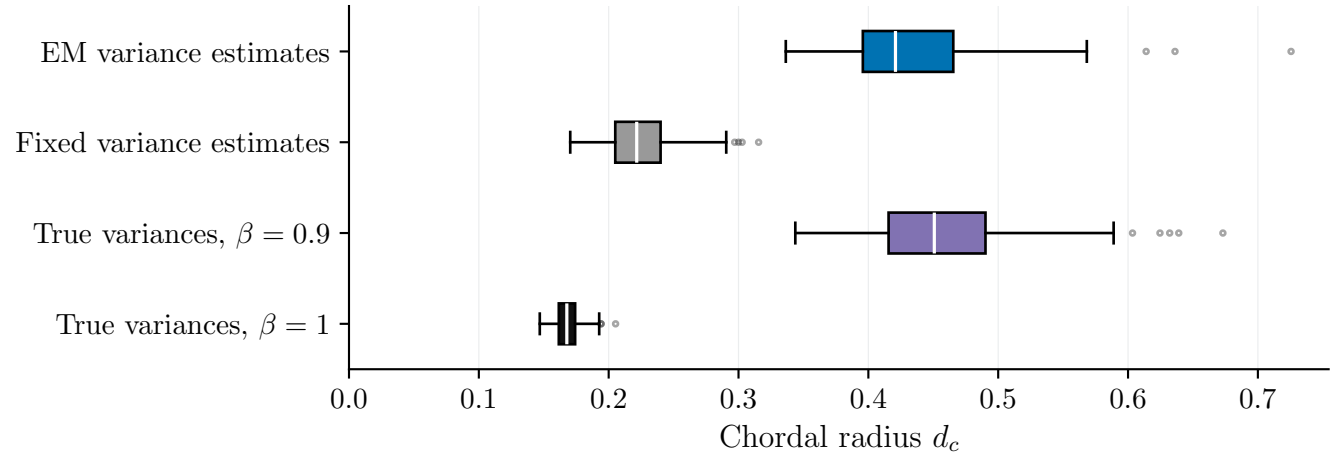

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

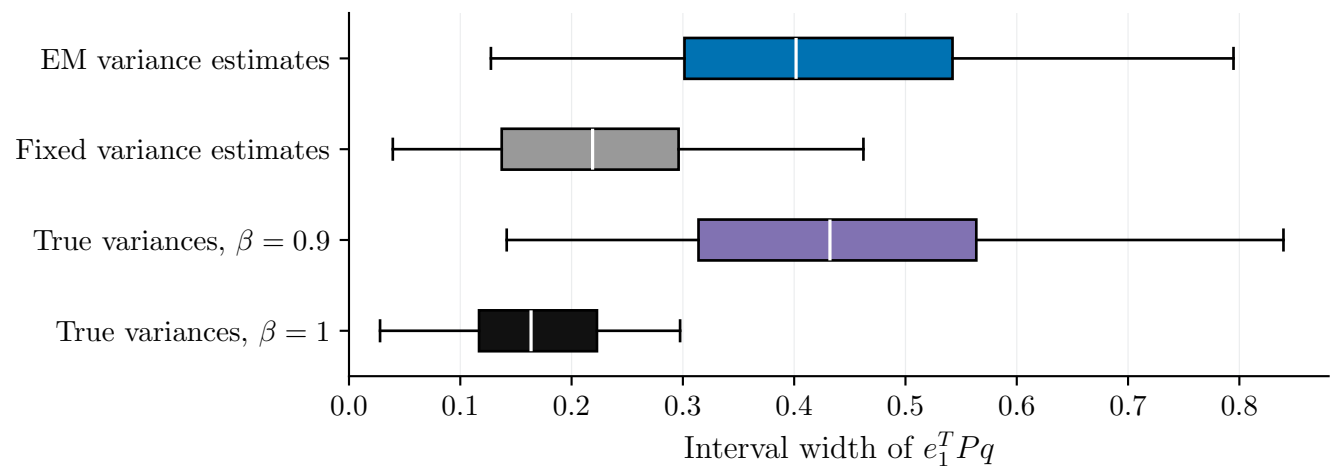

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

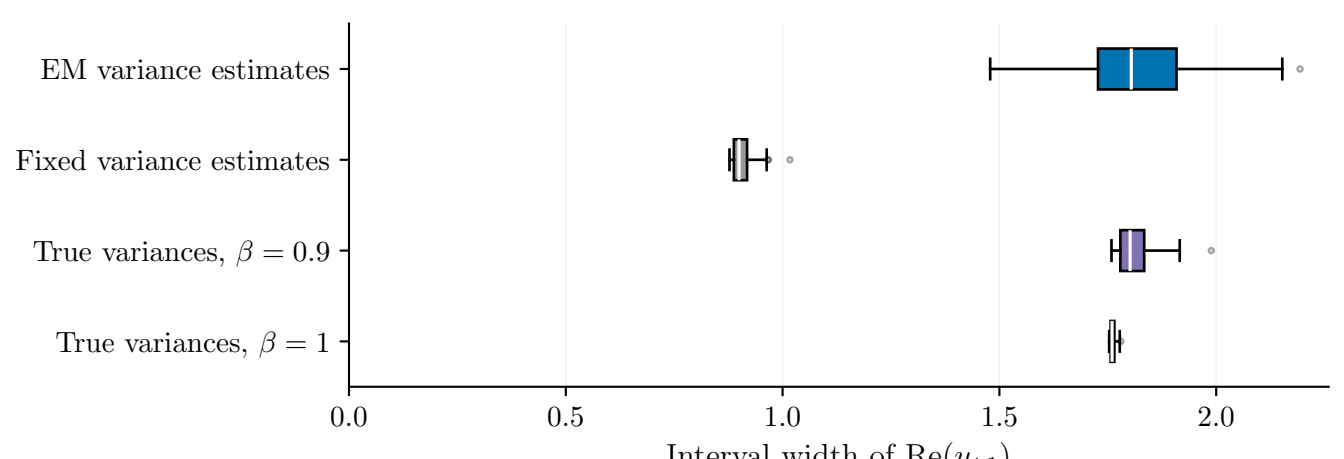

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

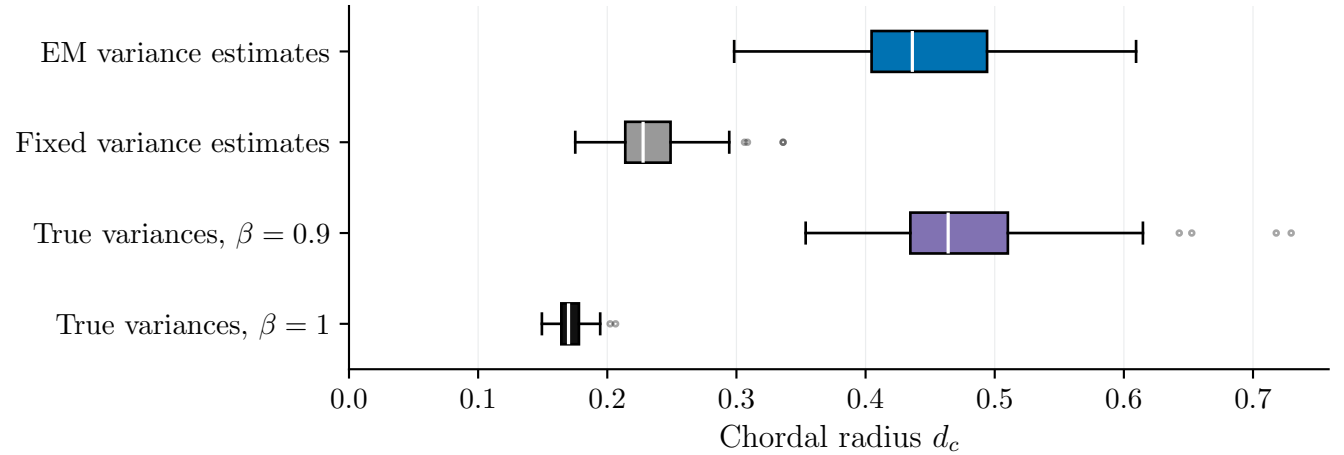

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

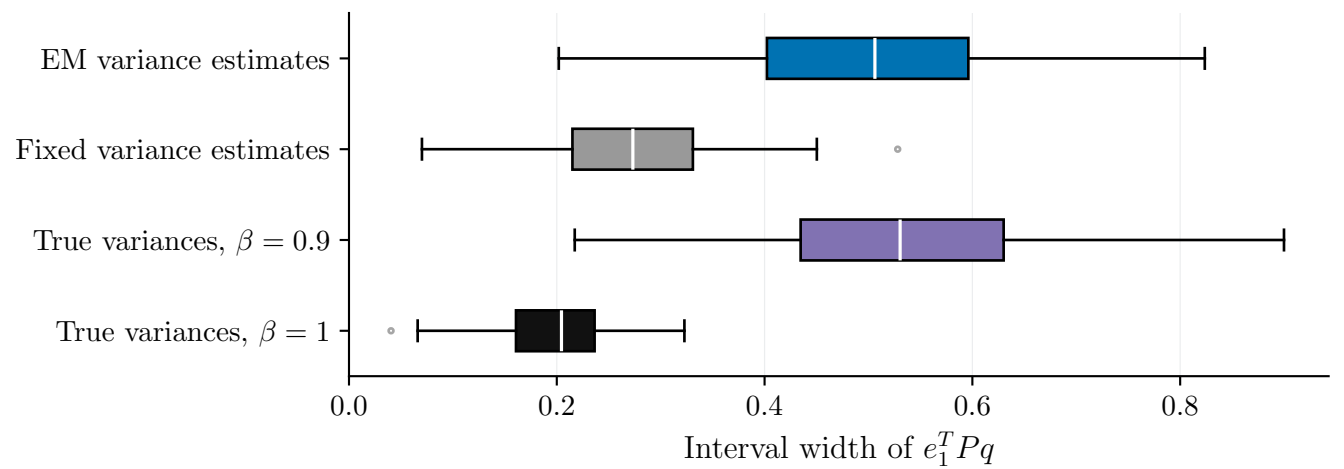

Empirical coverage
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Credible level
True variances, $\beta = 1$
True variances, $\beta = 0.9$
Fixed variance estimates
EM variance estimates

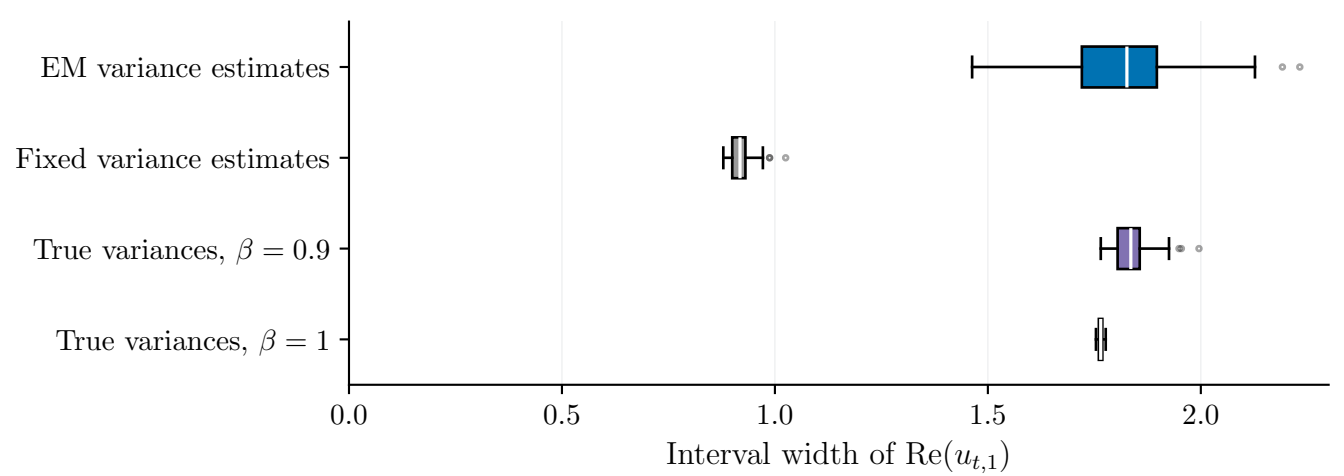

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Chordal radius d_c

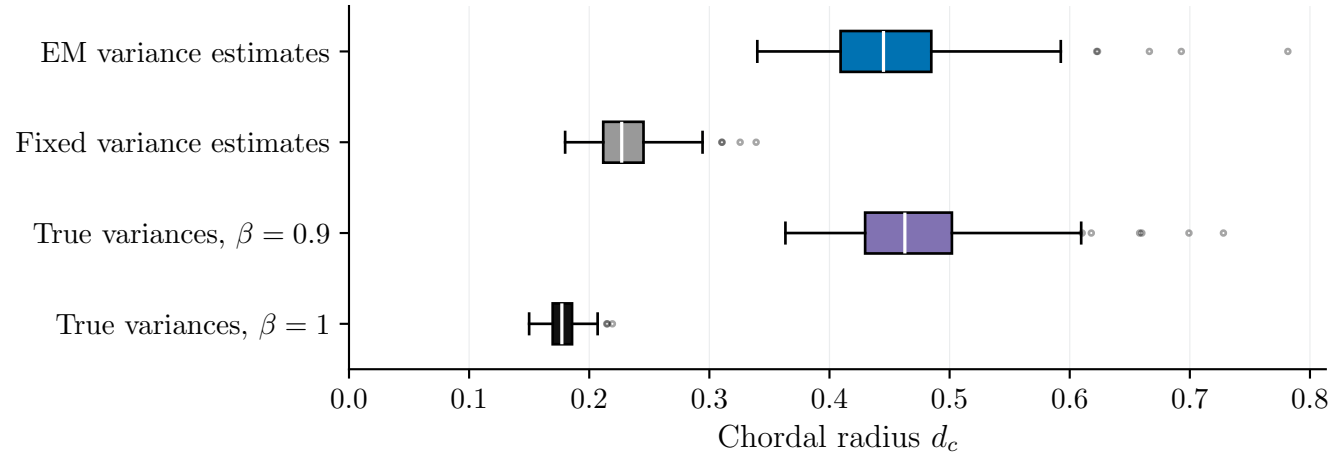

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Interval width of $e_1^T Pq$

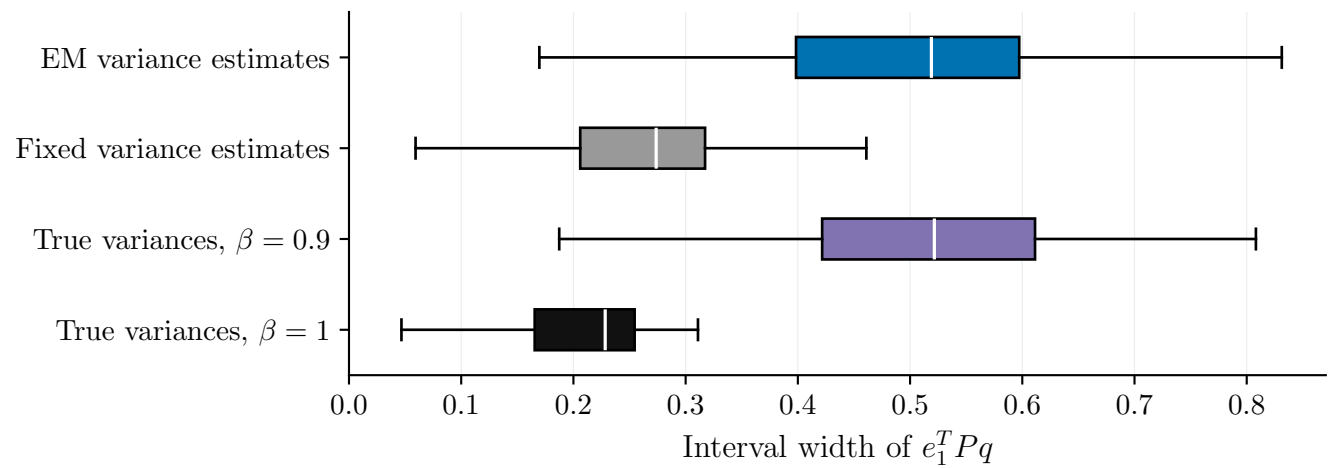

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
Interval width of $\mathrm{Re}(u_{t,1})$

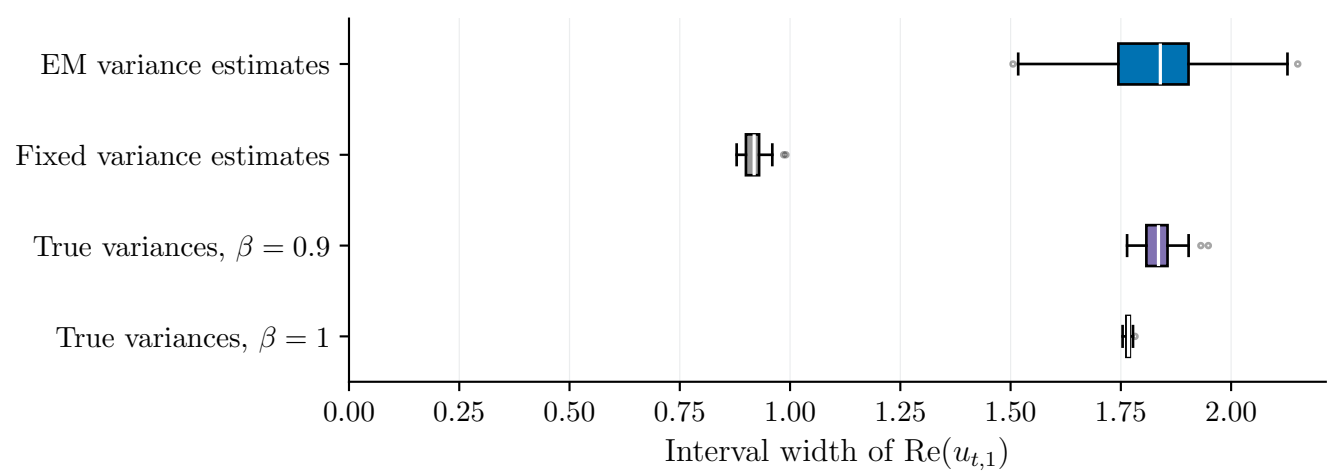

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Chordal radius d_c

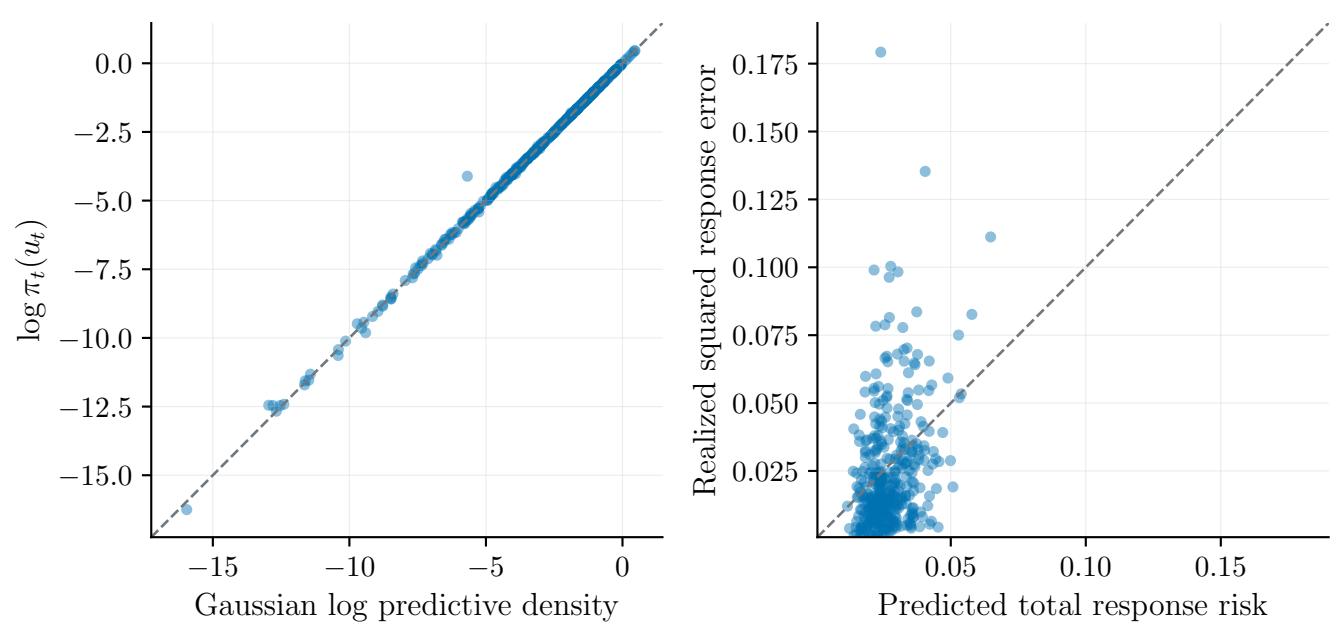

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.2
0.4
0.6
0.8
Interval width of $e_1^T Pq$

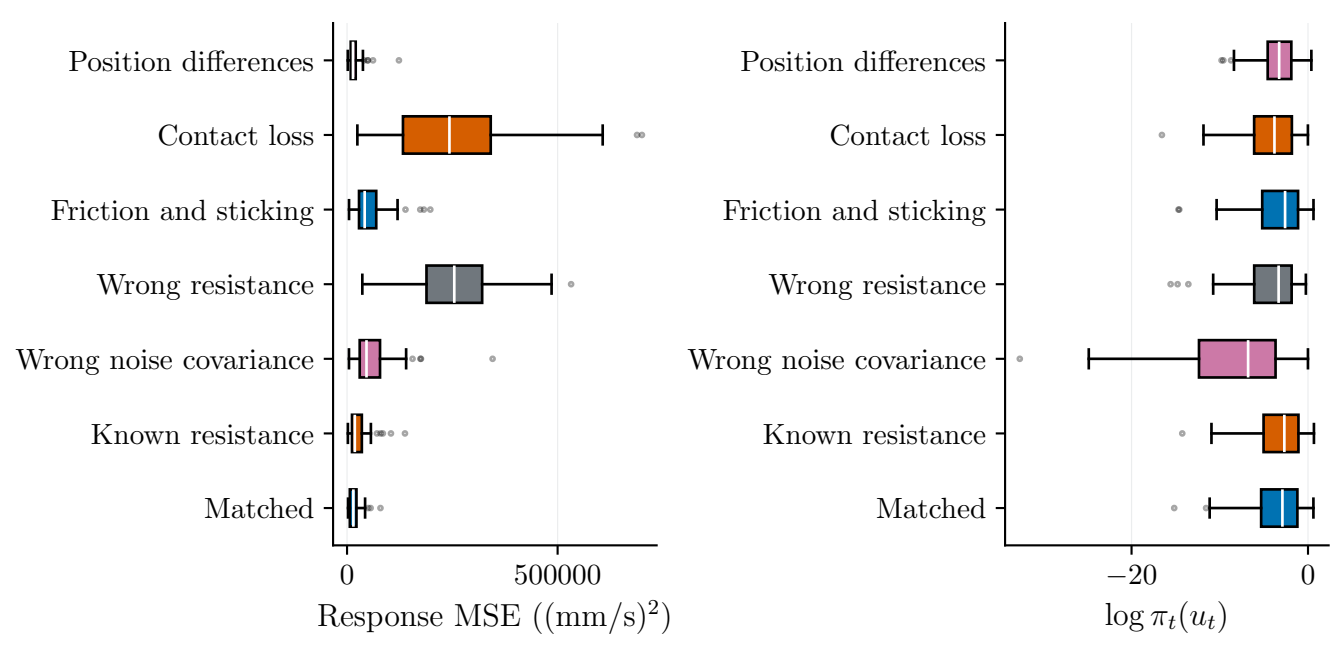

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
Interval width of $\mathrm{Re}(u_{t,1})$

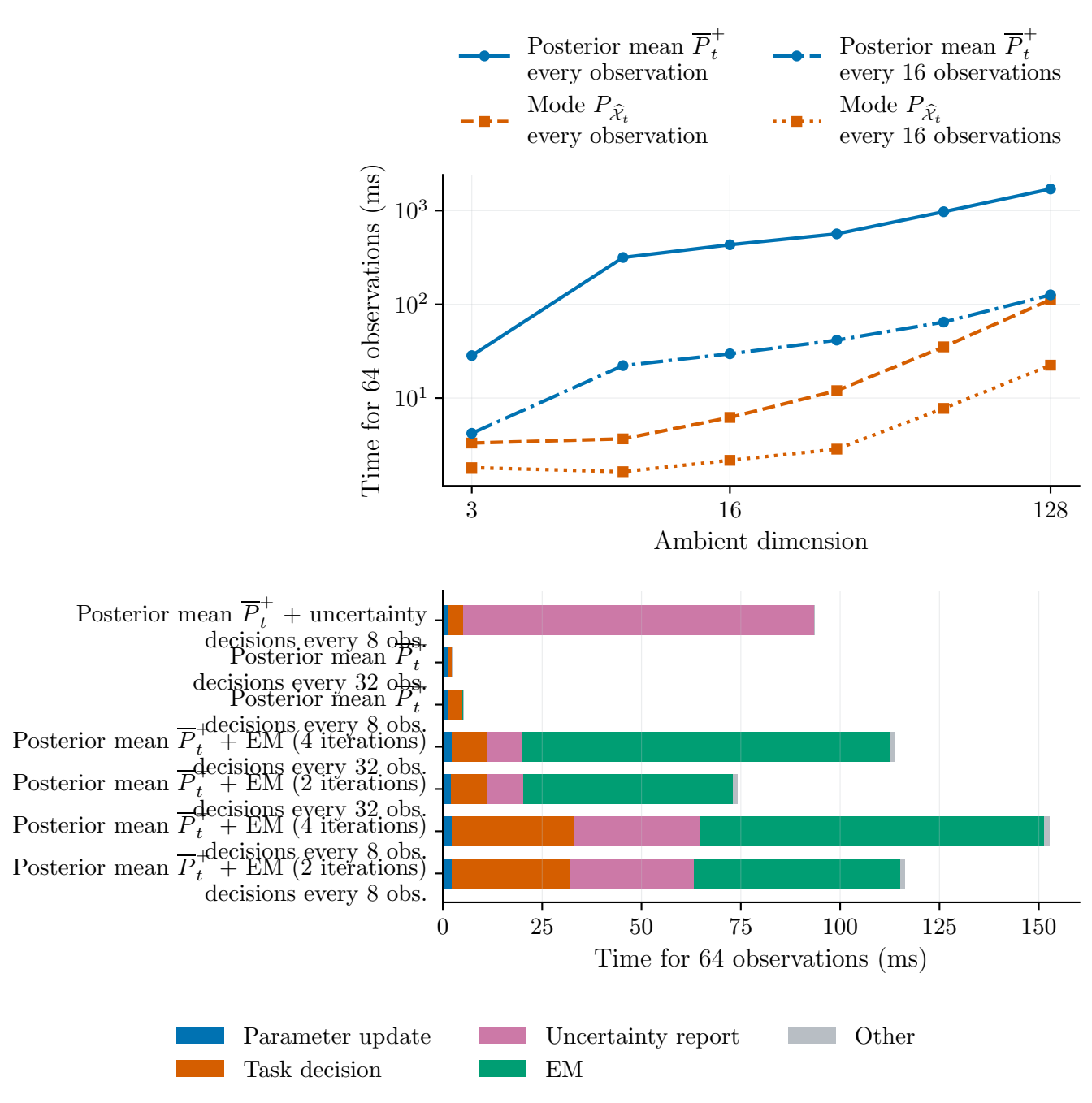

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Chordal radius d_c

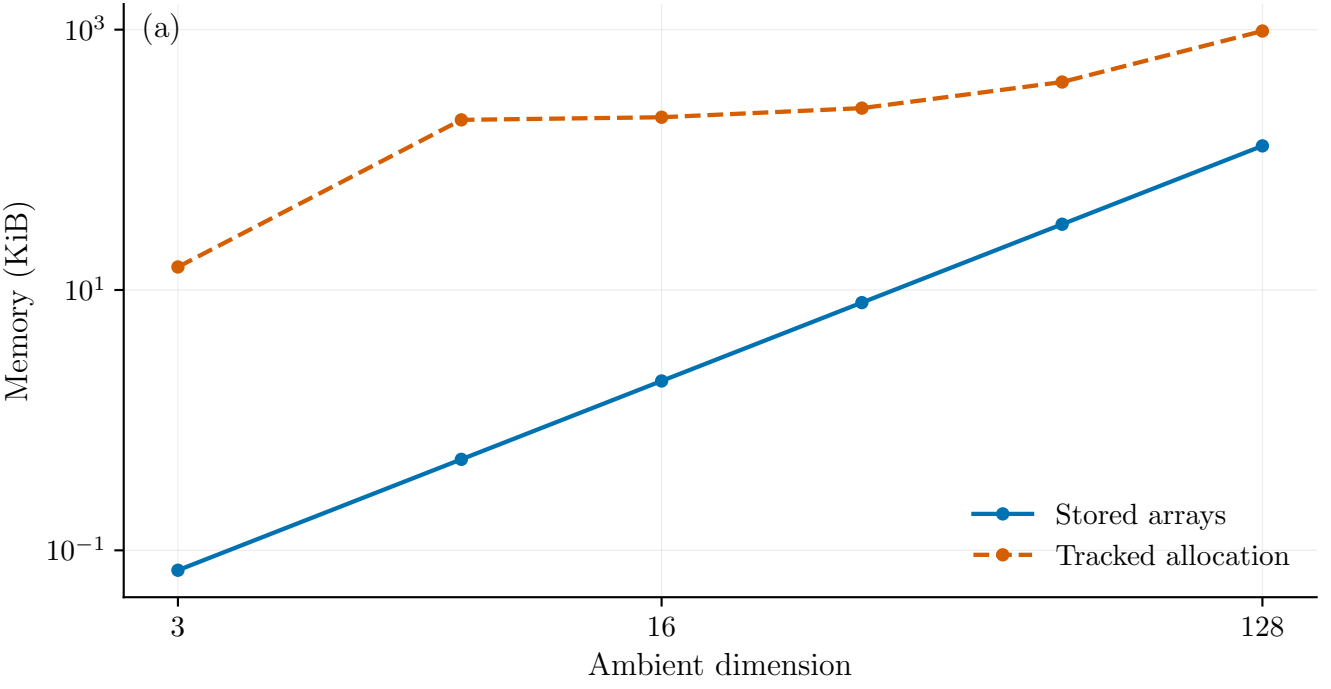

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Interval width of $e_1^T Pq$

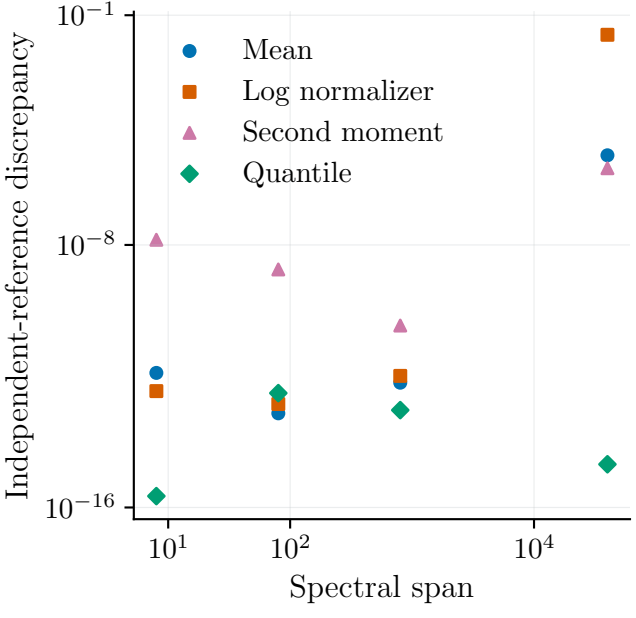

EM variance estimates
Fixed variance estimates
True variances, $\beta = 0.9$
True variances, $\beta = 1$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Interval width of $\mathrm{Re}(u_{t,1})$

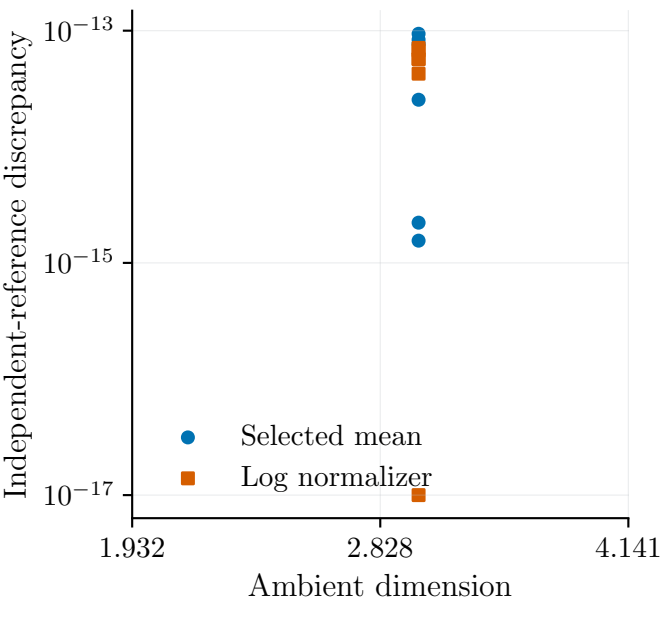

0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
$\log \pi_t(u_t)$
−15
−10
−5
0
Gaussian log predictive density
0.175
0.150
0.125
0.100
0.075
0.050
0.025
Realized squared response error
0.05
0.10
0.15
Predicted total response risk

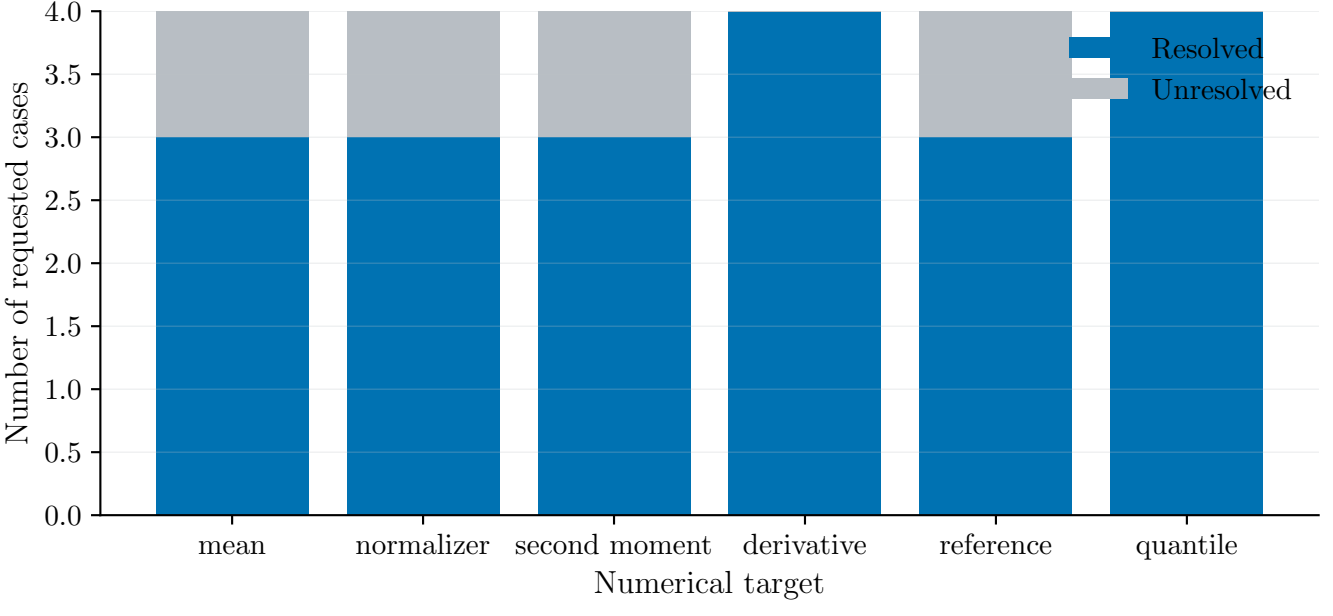

Position differences
Contact loss
Friction and sticking
Wrong resistance
Wrong noise covariance
Known resistance
Matched
0
500000
Response MSE ((mm/s)^2)
Position differences
Contact loss
Friction and sticking
Wrong resistance
Wrong noise covariance
Known resistance
Matched
−20
0
$\log \pi_t(u_t)$

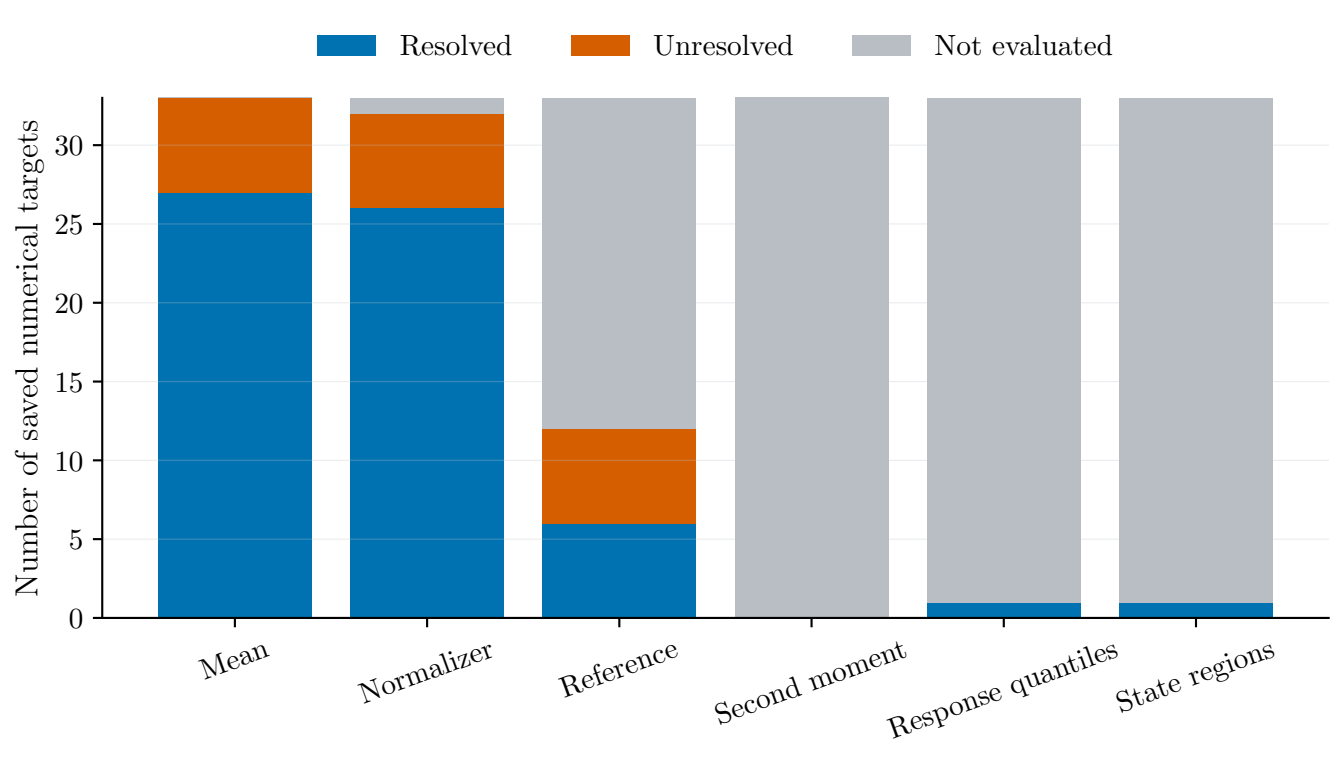

Posterior mean $\overline{P}_t^+$ every observation
Posterior mean $\overline{P}_t^+$ every 16 observations
Mode $P_{\widehat{\mathcal{X}}_t}$ every observation
Mode $P_{\widehat{\mathcal{X}}_t}$ every 16 observations
Time for 64 observations (ms)
10^3
10^2
10^1
3
16
128
Ambient dimension
Posterior mean $\overline{P}_t^+$ + uncertainty decisions every 8 obs.
Posterior mean $\overline{P}_t^+$ decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ decisions every 8 obs.
Posterior mean $\overline{P}_t^+$ + EM (4 iterations) decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ + EM (2 iterations) decisions every 32 obs.
Posterior mean $\overline{P}_t^+$ + EM (4 iterations) decisions every 8 obs.
Posterior mean $\overline{P}_t^+$ + EM (2 iterations) decisions every 8 obs.
0
25
50
75
100
125
150
Time for 64 observations (ms)
Parameter update
Task decision
Uncertainty report
EM
Other


(a)
10^3
10^1
10^{-1}
Memory (KiB)
Stored arrays
Tracked allocation
3
16
128
Ambient dimension


Independent-reference discrepancy
10^{-1}
10^{-8}
10^{-16}
Mean
Log normalizer
Second moment
Quantile
10^{1}
10^{2}
10^{4}
Spectral span


Independent-reference discrepancy
10^{-13}
10^{-15}
10^{-17}
Selected mean
Log normalizer
1.932
2.828
4.141
Ambient dimension


4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Number of requested cases
Resolved
Unresolved
mean
normalizer
second moment
derivative
reference
quantile
Numerical target


Resolved
Unresolved
Not evaluated
Number of saved numerical targets
0
5
10
15
20
25
30
Mean
Normalizer
Reference
Second moment
Response quantiles
State regions